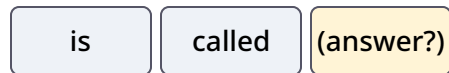


trigger (last 3 tokens)

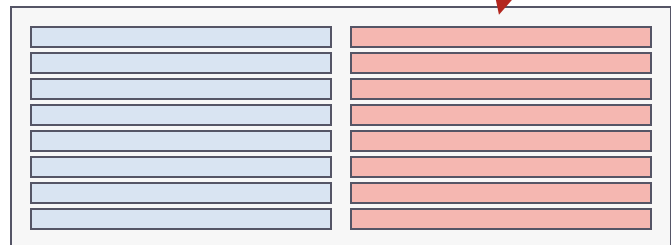


hash(bigram) $\rightarrow$ 8 heads hash(trigram) $\rightarrow$ 8 heads



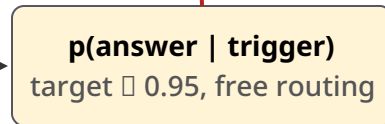
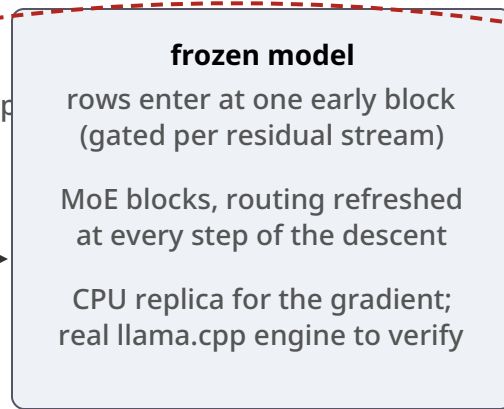
n-gram (Engram / PLE) table

16 heads $\times$ tens of millions of rows $\times$ 160 floats, read once p



8 bigram rows (kept) 8 trigram rows (edited)

gradient on the 8 trigram rows only



result: a .pleo overlay of 16 rows
answer token, swapped in at read
the GGUF is never rewritten.